Starhash: A Time-and-Place Seal Written as the Sky

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Abstract

Starhash draws the sky over a stated place at a stated moment and uses the drawing as a seal for a document. The sky carries the place and the time; a ring of ticks around the horizon carries the first 128 bits of the document's SHA-256. The seal holds no text and no hidden record. A reader recovers latitude, longitude and time from the geometry alone by identifying the stars and using the Moon, Sun and planets as clock hands. With the Moon above the horizon the time is recovered to seconds; with no Moon, Sun or planet above the horizon it cannot be recovered at all. The model is small and approximate, and a seal records a claim rather than proving it. This note describes the model, the seal, the reader and their limits.

1 Idea

A date and a pair of coordinates depend on conventions: a calendar, an epoch, a prime meridian. The sky over a place at a moment depends on much less. It can be drawn, and from the drawing the place and moment can be worked out again by anyone who knows the stars. Starhash is a small program built on that observation. It has three operations: *sign* draws a seal for a file, *read* recovers place and time from a seal, and *verify* checks a seal against a file. It is written in Python with no dependencies.

2 Model

The model is deliberately compact. Its ingredients are standard [2].

Time. Dates are converted to Julian Day in the proleptic Gregorian calendar, and $T = (JD - 2451545)/36525$. UTC is used directly as the time argument. Local sidereal time is the mean sidereal time polynomial at Greenwich plus the east longitude λ:

$$\theta = 280.46061837^\circ + 360.98564736629^\circ\, d + \lambda, \quad (1)$$

with $d = JD - 2451545$ and two small terms in T^2 and T^3.

Sun and planets. Mercury, Venus, Earth, Mars, Jupiter and Saturn move on ellipses whose elements change linearly with T [3]. Kepler's equation $M = E - e\sin E$ gives each heliocentric position; subtracting the Earth's gives the geocentric one, which is rotated by the mean obliquity into right ascension and declination.

Moon. Longitude, latitude and distance are sums of selected periodic terms of the ELP-2000 theory [1]. Phase and the orientation of the bright limb follow from the directions of Sun and Moon.

Observer. Positions of the Moon and planets are shifted for the observer's place on the WGS84 ellipsoid (diurnal parallax, up to 1° for the Moon). Altitudes are raised by Bennett's refraction formula at standard pressure and temperature.

Stars. The catalogue has 457 stars to magnitude 4 with J2000 positions, moved to the date by linear precession:

$$\Delta\alpha = (m + n\sin\alpha\tan\delta)\,T, \qquad \Delta\delta = n\cos\alpha\;T. \quad (2)$$

Accuracy. Against worked examples in [2] the Moon is within about $1'$ and the Sun and Venus within about $6'$. The model omits ΔT, UT1, nutation, aberration, light-time and proper motion. It is not a precision ephemeris.

3 Seal

Altitude a and azimuth A are mapped to a disc by the stereographic projection centred on the zenith,

$$r = \tan\Big(\frac{90^\circ - a}{2}\Big), \quad x = r\sin A, \quad y = -r\cos A, \quad (3)$$

so the horizon is the circle $r = 1$ and north is at the top.

Every mark carries information (Table 1); there is no text, so no font or alphabet is needed to draw or read a seal. The document enters as a ring of 128 ticks between the horizon and the rim. Read clockwise from north, a long tick is 1 and a short tick is 0, and the bits are the leading 128 bits of the file's SHA-256. Coordinates are rounded to five decimals and time to the

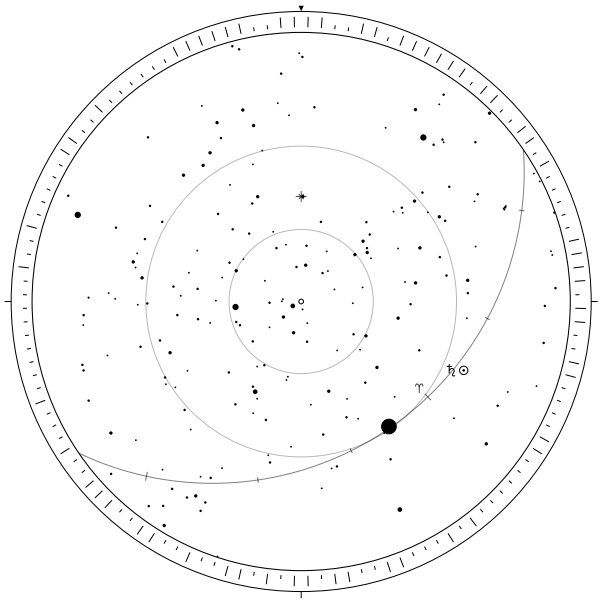


Figure 1: Seal of a file for Zürich (47.3769° N, 8.5417° E, 400 m) at 22:00 UTC on 28 August 2026. The Moon and Saturn are above the horizon.

Mark	Meaning
Heavy circle	horizon
Small central ring	zenith
Two faint circles	altitudes 60° and 30°
Wedge; three strokes	north; east, south, west
Dots, five sizes	stars by brightness
Eight-armed cross	celestial pole
Curved line	ecliptic
Strokes across it	the twelve signs, every 30°
Ram's horns	vernal point
Part-filled disc	Moon and its phase
Ringed point with sign	planet
Ring with centre point	Sun
Ticks in the rim	document, 128 bits

Table 1: The marks of a seal.

second before drawing, so equal inputs give identical seals.

4 Reading

The reader uses only the geometry of the drawing.

Stars. Each dot is taken back through the projection and the refraction to a direction above the horizon. The stars carry no labels. Two bright dots are matched to a pair of catalogue stars of the same brightness class and the same angular separation; this fixes a trial orientation of the sky, which is kept if it brings the other bright dots onto catalogue stars. A least-squares rotation over all matched stars refines it. The direction of the zenith among the stars gives the latitude and the local sidereal time.

Fastest body up	Share	Time	Longitude
Moon	49%	6 s	0.02°
Sun or Mercury–Mars	38%	44 s	0.19°
Jupiter, Saturn only	8%	20 min	5°
None	4%	—	—

Table 2: Median error of time and longitude read back from 300 random seals, by the fastest body above the horizon: the Moon; else the Sun, Mercury, Venus or Mars; else Jupiter or Saturn.

Clock hands. Sidereal time alone does not give the date. The positions of the Sun, Moon and planets among the stars do, and they do not depend on the observer's longitude. The reader scans a stated interval of years (1800 to 2200 by default) for the moments when the model puts the bodies where they are drawn. Planets are recognised by their signs.

Fit. From each candidate moment, latitude φ, longitude λ and Julian Day are adjusted together to minimise

$$\sum_i \left\| p_i(\varphi, \lambda, JD) - y_i \right\|^2 \tag{4}$$

over the drawn bodies and the brightest stars, where p_i is the model position and y_i the drawn one. The best candidate is returned with an uncertainty estimated from the resolution of the drawing.

Verification. To verify a seal, the reader's result is drawn again and every star and body in the seal must coincide with one in the new drawing. The ticks are compared with the hash of the file offered. If a place and time are stated, the seal is also compared with the seal of exactly those values.

5 Results

Table 2 summarises 300 seals drawn for random places between 85° S and 85° N and random moments between 1800 and 2200, at the default size of 512 units, and then read back. Every seal that showed at least one body was assigned to the right date; one, showing Saturn alone near its retrograde loop, fitted a second date equally well, and the reader listed both. Latitude was recovered with a median error of 0.00014° (15 m) and a largest error of 0.0005° (60 m).

How sharply a seal fixes the time depends on its fastest hand. The Moon moves about half a degree an hour and gives seconds. Jupiter and Saturn barely move in a day, and near their stationary points a seal showing only one of them leaves hours of doubt, which the reader reports. The largest errors in that row were 11 hours and, for longitude, the whole circle. Longitude follows from the time at 15° per hour, so it inherits the same doubt. These figures measure the reader against the model that drew the seal. Read against the true sky, the errors of the model in Section 2 would be added.

6 Limits

A seal is a claim. The sky of any place and moment can be computed by anyone, so a seal cannot show that its maker was there then. Verification shows that the drawing is a consistent sky and that the document is the one in the ring. Trust in the claim has to come from elsewhere, for example a cryptographic signature or a public timestamp over the seal.

Some skies are silent. With no Moon, Sun or planet above the horizon a seal gives latitude and sidereal time only. Marking the bodies below the horizon would remove this gap and is not yet done.

Conventions remain. Longitude is counted from Greenwich, the planets are marked with traditional signs, and the reader must be given the interval of years to search and the star catalogue. The altitude circles, the pole and the signs on the ecliptic make the construction legible, but the seal does not yet carry its own legend.

Scope. The reader decodes the program's own vector drawing. Reading a photograph or scan of a printed seal is not implemented. The model is meant for seals, not for navigation, surveying or historical dating.

References

[1] Michelle Chapront-Touzé and Jean Chapront. The lunar ephemeris elp 2000. *Astronomy and Astrophysics*, 124:50–62, 1983.

[2] Jean Meeus. *Astronomical Algorithms*. Willmann-Bell, Richmond, VA, 1998.

[3] JL Simon, P Bretagnon, J Chapront, M Chapront-Touze, G Francou, and J Laskar. Numerical expressions for precession formulae and mean elements for the moon and the planets. *Astronomy and Astrophysics*, 282:663–683, 1994.